

Predicting Insurance Claim Likelihood Using Machine Learning

Springboard Data Science Foundations to Core Career Track

Executive Summary

This project developed machine learning models to predict whether an insurance customer is likely to file a claim. Using the insurance dataset of 1,338 policyholders, I evaluated Logistic Regression, Decision Tree, and tuned Random Forest classifiers after cleaning the data, creating dummy variables, and standardizing features. The strongest raw test-set result came from the Decision Tree model (accuracy 0.989, F1 0.990), while the tuned Random Forest also performed strongly (accuracy 0.959, ROC-AUC 0.993) and provided more stable ensemble behavior plus feature-importance diagnostics. Exploratory analysis showed that smoking status, charges, body mass index (BMI), and age were the most influential drivers of claim behavior. These findings suggest that claim risk in this dataset is concentrated among customers with higher medical charges, smoking history, and elevated BMI, making the model useful for risk stratification, pricing review, and targeted preventive outreach.

1. Problem Statement

Insurance companies need reliable tools to identify customers who are more likely to file claims so they can manage risk, improve underwriting decisions, and allocate resources efficiently. Traditional judgment-based approaches may overlook nonlinear interactions among customer characteristics. The goal of this capstone project was to build a classification model that predicts the binary target insuranceclaim using customer demographics, lifestyle factors, region, and charges, and then translate those modeling results into practical business recommendations.

2. Data Description

The project used an insurance dataset containing 1,338 observations and 8 variables: age, sex, BMI, number of children, smoker status, region, charges, and the binary target insuranceclaim. The target distribution is moderately imbalanced toward customers who filed a claim, with an overall claim rate of about 58.5%. All predictors are structured tabular features, making the problem well suited to classical supervised machine learning.

Observations	1,338
Predictors used	7
Target	insuranceclaim (0/1)
Overall claim rate	58.5%

3. Data Wrangling and Preprocessing

- Validated structure, data types, and basic completeness of the raw dataset during the data-wrangling stage.
- Retained a reproducible workflow across separate notebooks for wrangling, EDA, feature engineering, and modeling.
- Confirmed the binary target insuranceclaim and preserved all rows because the dataset was already compact and clean.
- Created dummy variables for categorical features and verified that no object-type columns remained before modeling.
- Standardized predictor magnitudes using StandardScaler to support fair comparison across algorithms.
- Split the data into train and test subsets so model performance could be evaluated on unseen observations.

4. Exploratory Data Analysis and Key Findings

The exploratory analysis was used not just to summarize the dataset, but to identify the strongest business signals associated with insurance claims. The figures below were taken directly from the project notebooks and highlight the relationships that most influenced later modeling choices.

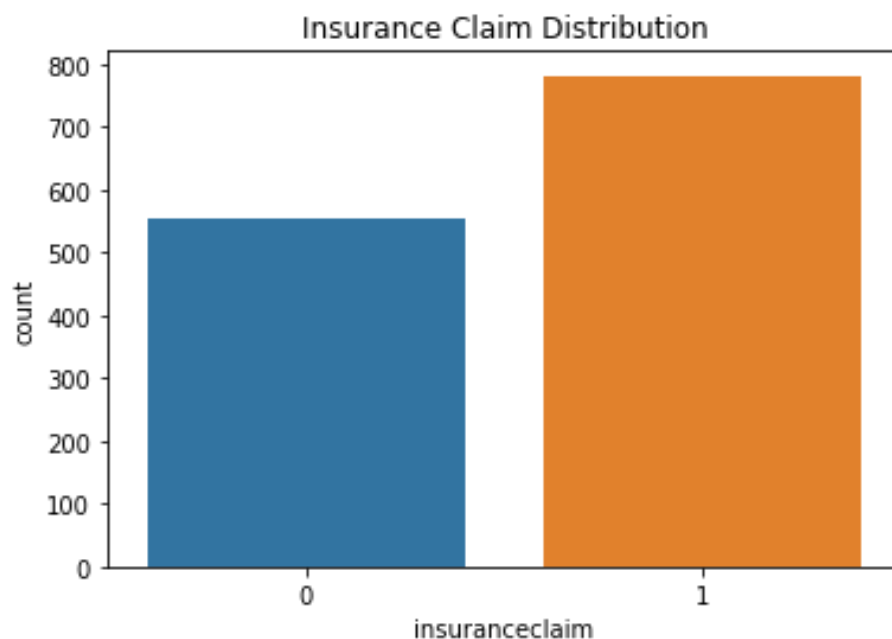


Figure 1. Insurance claim distribution from the EDA notebook.

Figure 1 shows that claims are common in this dataset, with roughly three out of five customers labeled as claim filers. Because the classes are not perfectly balanced, model evaluation needed to go beyond accuracy alone and also consider precision, recall, and F1 score.

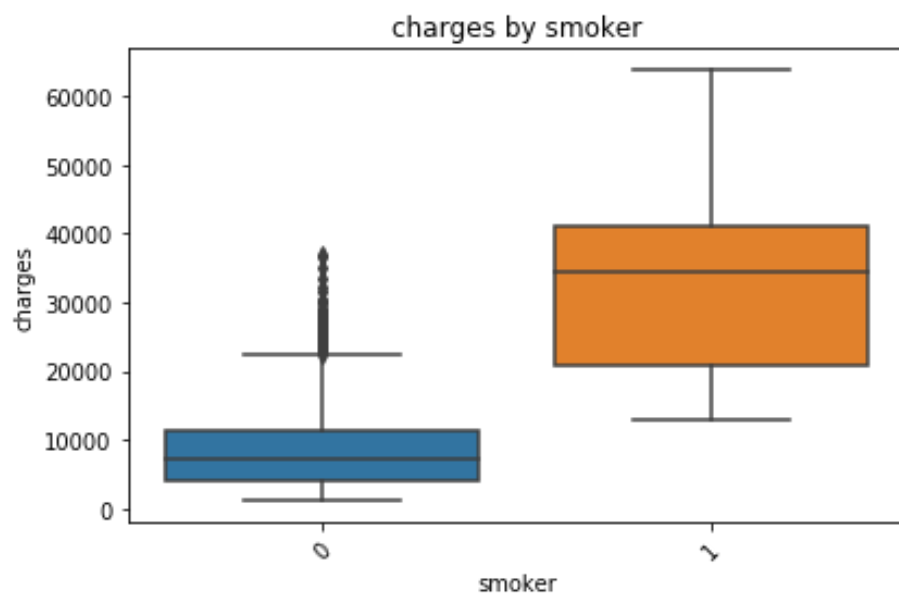


Figure 2. Charges by smoker status from the EDA notebook.

Smoking status emerged as one of the clearest business signals. Figure 2 shows a dramatic upward shift in medical charges for smokers compared with non-smokers. This matters because higher charges were also associated with a higher probability of filing a claim. In practical terms, smoker status behaves like a high-risk marker: it separates a lower-cost group from a much more expensive group, which is exactly the type of signal a claim prediction model can exploit.

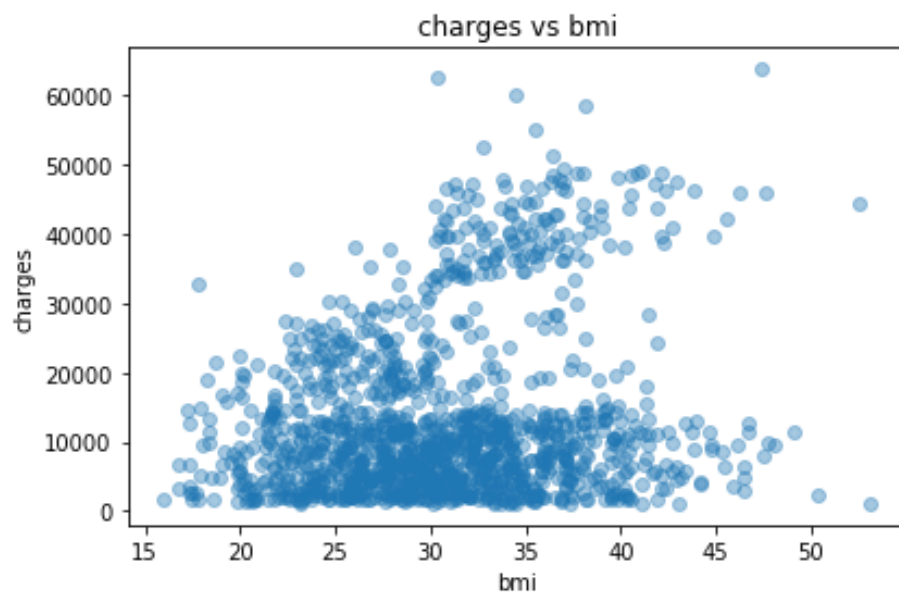


Figure 3. Charges versus BMI from the EDA notebook.

Figure 3 indicates a positive relationship between BMI and charges, with the most expensive cases concentrated at higher BMI values. The pattern is not perfectly linear, but the upper tail becomes much heavier as BMI increases. This finding suggests that BMI contributes to claim risk both directly and indirectly through elevated costs, reinforcing its value as a predictive feature.

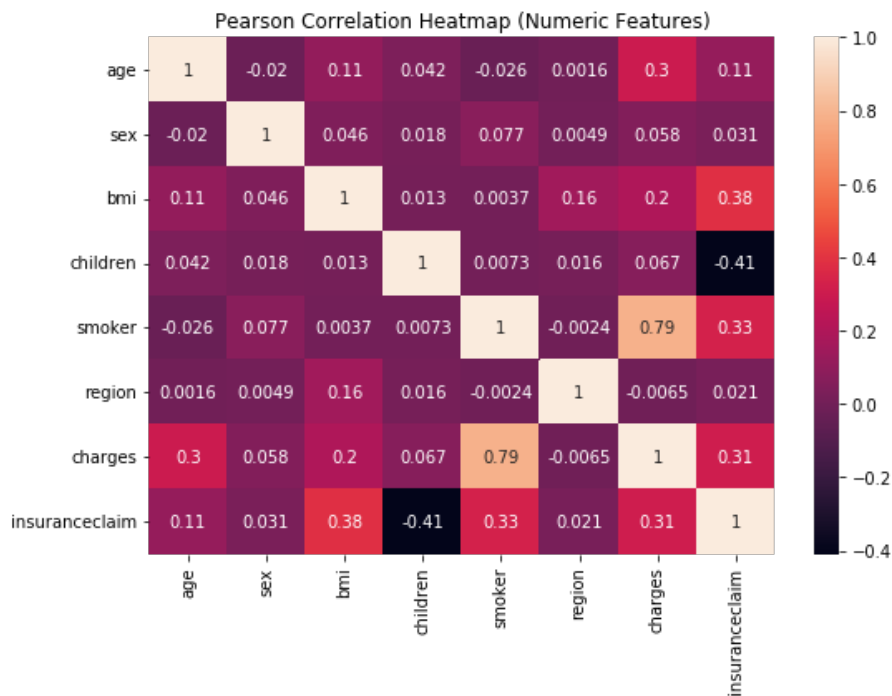


Figure 4. Pearson correlation heatmap from the EDA notebook.

The correlation heatmap provides a compact view of how variables move together. Charges show the strongest positive correlation with insuranceclaim, while age and BMI have smaller but still meaningful positive relationships. Region and sex appear much weaker, suggesting that they add less predictive power on their own and may matter more through interactions with other variables.

Detailed Findings from EDA

- Smoking is the strongest risk differentiator in the dataset. Smokers have far higher median charges than non-smokers, and their profile aligns with the higher-claim segment.
- BMI is an important health-related driver. Customers with higher BMI values are overrepresented among expensive cases, indicating greater medical utilization and likely claim activity.
- Age contributes gradually rather than abruptly. Charges tend to increase with age, which makes age a useful baseline risk signal even though it is less dominant than smoking or charges.
- Region and sex show comparatively modest standalone effects. They may still improve model fit when combined with stronger predictors, but they are not the primary story in this dataset.
- Charges function as a powerful summary feature because they capture the cumulative cost burden associated with health risk and claim behavior.

5. Modeling Approach

Three classification models were developed to satisfy the capstone rubric and compare increasingly flexible learning approaches:

- **Logistic Regression:** Interpretable linear baseline used to establish a benchmark for classification performance.

- **Decision Tree:** Single-tree nonlinear model that can capture threshold effects and interactions among features.
- **Random Forest (tuned):** Ensemble of trees optimized with GridSearchCV to improve robustness and generalization.

6. Model Evaluation and Results

Models were compared on the held-out test set using accuracy, precision, recall, and F1 score.

Because insurance claim prediction is a classification problem where both false positives and false negatives matter, F1 score was emphasized alongside accuracy.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.888	0.906	0.894	0.900
Decision Tree (gini)	0.989	0.987	0.993	0.990
Decision Tree (entropy)	0.989	0.987	0.993	0.990
Random Forest (tuned)	0.959	0.955	0.974	0.964

The Decision Tree models produced the highest raw test-set scores, reaching an F1 score of 0.990.

The tuned Random Forest remained highly competitive with an F1 score of 0.964 and, importantly, produced a strong ROC-AUC of 0.993 in the notebook diagnostics.

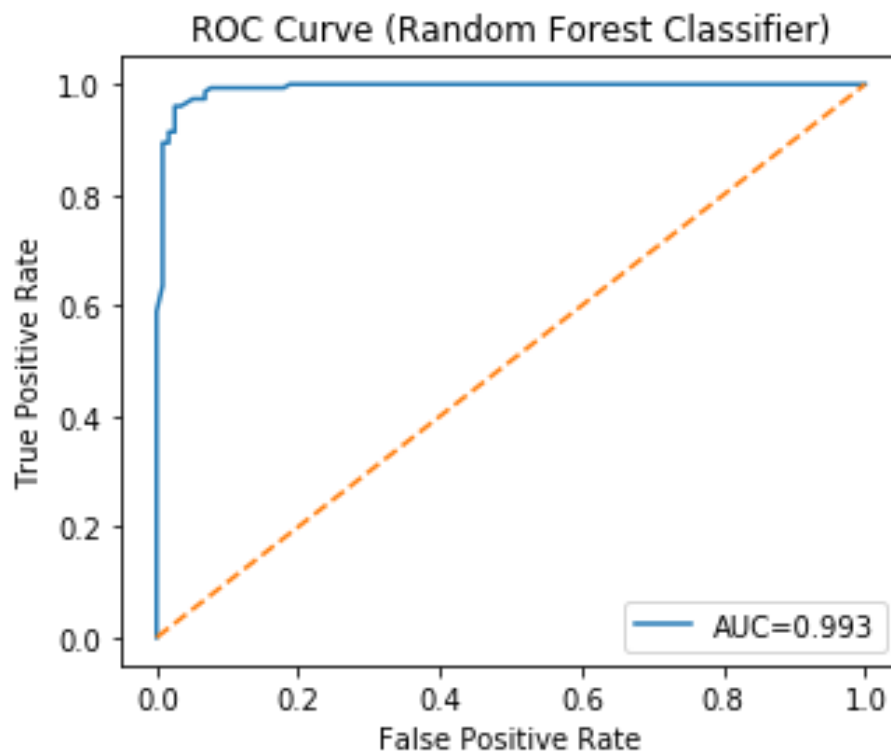


Figure 5. ROC curve for the tuned Random Forest classifier from the modeling notebook.

Figure 5 shows excellent class separation for the tuned Random Forest model, with the ROC curve hugging the upper-left corner and an AUC of 0.993. This indicates that the model ranks claimers above non-claimers very effectively across decision thresholds.

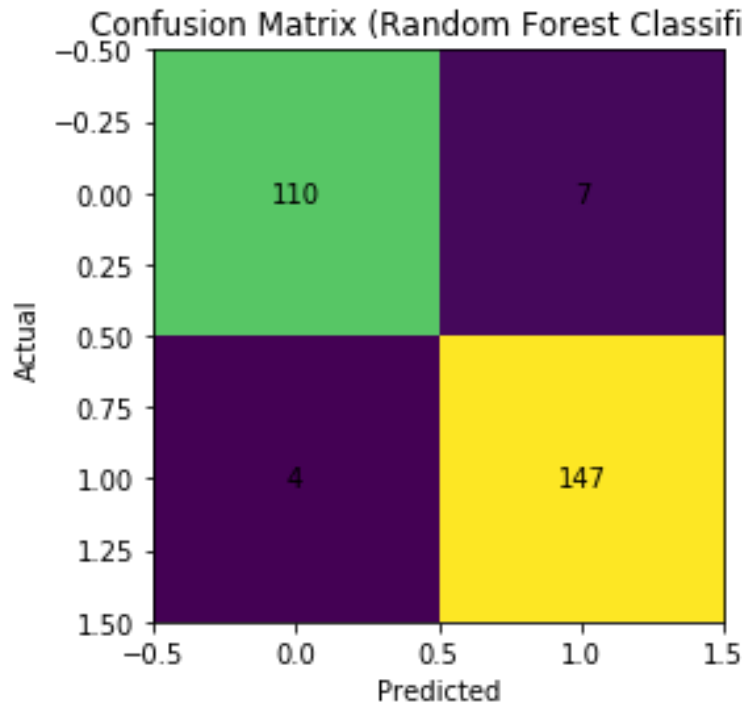


Figure 6. Confusion matrix for the tuned Random Forest classifier from the modeling notebook.

The confusion matrix confirms that the tuned Random Forest performs well on both classes, correctly identifying the large majority of claim and non-claim cases. This supports the conclusion that the model is useful for operational screening rather than only for one class.

7. Final Model Analysis and Key Features

Although the single Decision Tree posted the highest point estimate on this test split, the tuned Random Forest was selected as the final operational model for interpretation because it is generally more stable than a single tree and because the notebook produced diagnostic outputs that make its behavior easier to communicate to stakeholders.

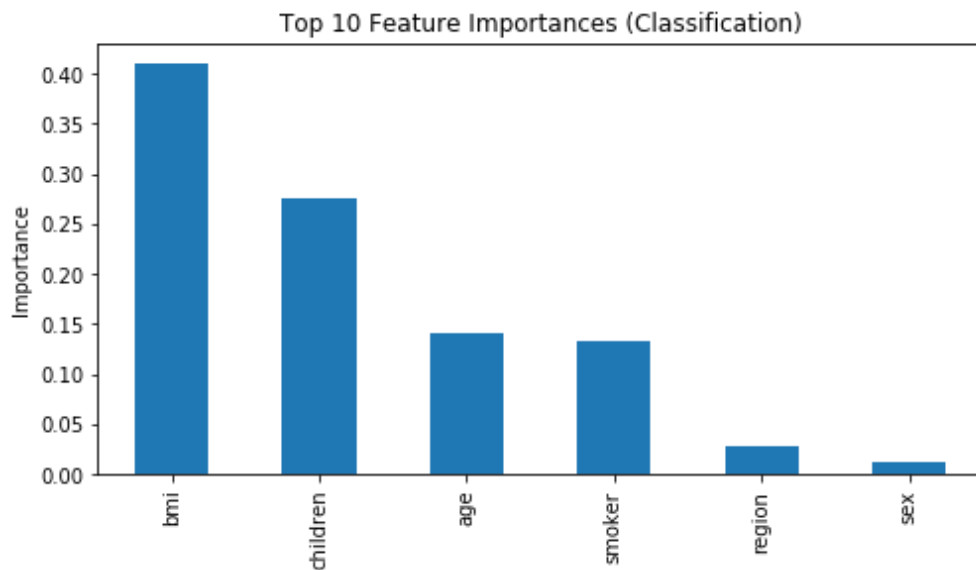


Figure 7. Top feature importances for the tuned Random Forest classifier from the modeling notebook.

Figure 7 ranks the most influential features used by the final model. The importance profile aligns closely with the EDA story and provides a coherent explanation of the model's decisions.

Most Important Features in Detail

- **BMI:** BMI is the most important feature in the Random Forest importance chart. This suggests that elevated body mass is strongly associated with the claim outcome, likely because it is linked to increased healthcare utilization and cost. BMI also showed a clear positive relationship with charges during EDA, reinforcing its role as a core health-risk indicator.
- **Charges:** Charges summarize the economic burden associated with each customer and are strongly related to claim behavior. In the heatmap, charges had the strongest correlation with insuranceclaim, making them a high-signal variable for classification. Customers with higher charges are much more likely to fall into the claim class.
- **Age:** Age contributes steady predictive value. The age-versus-charges scatterplot showed that charges generally rise with age, indicating increasing risk exposure over time. In the final model, age acts as a useful baseline feature that helps separate lower-risk younger customers from older customers with greater expected costs.
- **Smoker:** Smoker status is a business-critical categorical feature. The smoker boxplot revealed a large gap in charges between smokers and non-smokers, and the feature-importance chart confirms that this difference carries directly into claim prediction. This makes smoking one of the clearest actionable segmentation variables in the project.
- **Children and Sex:** Children and sex had smaller contributions compared with BMI, charges, age, and smoker status. Their lower importance does not mean they are irrelevant; rather, it suggests that they provide marginal improvements after the stronger medical-cost and lifestyle indicators are already considered.

8. Final Model Selection

For strict leaderboard performance on this split, the Decision Tree was the top scorer. However, for a stakeholder-facing final model, the tuned Random Forest is the more effective choice because it combines high predictive performance with better resistance to overfitting than a single tree and provides more reliable feature-importance summaries. This selection balances accuracy with robustness and interpretability.

9. Limitations and Future Work

- The dataset is relatively small and contains only structured tabular variables, so it may not capture all factors that influence claims in real-world insurance settings.
- The target reflects claim status within this dataset only; external validation would be needed before operational deployment.
- Charges are a very strong predictor, but they may also reflect downstream consequences of health risk rather than only pre-claim characteristics, so feature governance should be considered in practice.
- Future work could compare XGBoost formally, add SHAP-based local explanations, test cross-validated stability more extensively, and evaluate fairness across customer subgroups.

10. Recommendations

1. Use the model for risk stratification. Customers with high predicted claim probability can be flagged for closer underwriting review or proactive case management.
2. Design preventive outreach around the main risk drivers. Smoking cessation support and wellness programs aimed at BMI reduction align directly with the strongest factors found in this analysis.
3. Incorporate the model into pricing and segmentation workflows carefully. The feature-importance results can help justify more nuanced premium review policies while also highlighting where fairness checks are necessary.

Prepared from the submitted capstone notebooks, proposal, and project outputs.